

Controlled Formal Awareness Security

Awareness, Capability, Preorders, Federation, Proof Depth,
and Nonstandard Bounds for Intelligent Target Seekers

Working Draft

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Abstract

This working draft develops a security theory around *controlled formal awareness* for Intelligent Target Seekers (ITSs), a term used here to include both automated theorem provers and broader intelligent theorem settlers such as AMLD-style systems. The primitive quantity is the directed two-input relation $\text{awa}(M, N)$: the formal, verifier-compatible awareness that machine M has of machine N . Formal self-awareness is recovered as the diagonal $\text{lev}(M) = \text{awa}(M, M)$. The framework then introduces awareness matrices and directed graphs, target-relative preorders and induced equivalence classes, finite-budget awareness, awareness horizons, federation constraints, security-capability preorders, and a distinction between awareness depth and Frozen-Ocean proof depth. Nonstandard analysis is used only to supply ideal upper comparison bounds for finite behavior, not to assert that a deployed ITS literally becomes infinite or hyperfinite. The central security principle is that authorized awareness should be sufficient for function while unauthorized cross-machine awareness remains controlled.

1 Intelligent Target Seekers and Formal Awareness

Let an **Intelligent Target Seeker (ITS)** mean a formal machine whose purpose is to seek settlement of a target. This terminology includes both automated theorem provers and more general AMLD-style intelligent theorem settlers. The broader settlement architecture motivating this terminology is developed in Tenneson’s work on automated logical deciders and later DATA-MIND systems [6, 7].

Let $\mathcal{M}$ denote a class of ITSs. The primitive awareness relation is the directed two-input quantity

$$\boxed{\text{awa}(M, N)}, \quad M, N \in \mathcal{M}.$$

It measures the depth to which M has a verifier-compatible formal representation of N adequate for sound reasoning about N ’s relevant states, transitions, proofs, settlement behavior, memory, strategy, or reflective behavior.

Awareness is formal, not phenomenal. No claim about consciousness or subjective experience is

intended. The earlier self-awareness quantity is recovered on the diagonal:

$$\boxed{\text{lev}(M) := \text{awa}(M, M).}$$

Thus formal self-awareness is a special case of directed awareness. This extends the formalized self-modeling program developed in earlier work on induction, ML-guided proof search, and formalized self-awareness [3].

In general,

$$\text{awa}(M, N) \neq \text{awa}(N, M).$$

The relation is therefore naturally asymmetric.

2 Finite Awareness and the Meaning of ω

The operational quantity should be budgeted awareness:

$$\text{awa}_B(M, N),$$

the greatest finite reflective depth that M can realize about N under finite resource budget B . On the diagonal,

$$\text{lev}_B(M) = \text{awa}_B(M, M).$$

The unbudgeted quantity is best interpreted as a bound:

$$\boxed{\text{awa}(M, N) = \sup_B \text{awa}_B(M, N).}$$

Likewise,

$$\boxed{\text{lev}(M) = \sup_B \text{lev}_B(M).}$$

Accordingly,

$$\text{awa}(M, N) = \omega$$

does *not* mean that an ordinary ITS literally reaches an infinite awareness state. It means

$$\forall k < \omega \exists B < \infty \quad \text{awa}_B(M, N) \geq k.$$

Every requested finite reflective depth is attainable at some sufficiently large finite budget, but no particular finite run is required to attain ω .

Similarly,

$$\text{lev}(M) = \omega$$

means that the achievable finite self-awareness depths of M have no finite upper bound:

$$\forall k < \omega \exists B < \infty \quad \text{lev}_B(M) \geq k.$$

Thus ω is a supremum or upper-bound object for the finite hierarchy.

At the opposite endpoint,

$$\text{awa}(M, N) = 0$$

means that M has no verifier-compatible formal representation of N sufficient even for the first nontrivial level of target-specific reasoning. Merely storing N 's name, code, or uninterpreted output need not count as awareness.

3 Awareness of Submachines

Suppose M consists of submachines

$$S_1, \dots, S_r.$$

There is no fundamental reason that M cannot formally model each of them. Define

$$\text{SubAwa}_B(M) = \inf_{S \in \text{Sub}(M)} \text{awa}_B(M, S).$$

An architecture can therefore aim at high parent-to-component awareness, with $\text{awa}(M, S_i)$ large for every monitored S_i .

However, awareness of every component separately does not automatically provide complete awareness of the whole system. M must also model the **composition law**: communication edges, BANK dependencies, shared state, scheduling, verifier interaction, controller behavior, and emergent interactions among components.

Thus

$$\boxed{\text{awareness of parts} + \text{awareness of composition}}$$

is a better foundation for strong self-awareness than component awareness alone.

This observation introduces a breadth dimension distinct from reflective depth. The quantity lev should continue to measure depth, while the number or family of components and subsystem combinations that can be modeled measures awareness breadth.

4 The Awareness Matrix and Directed Graph

For machines

$$M_1, \dots, M_r,$$

define the awareness matrix

$$\boxed{A_{ij} = \text{awa}(M_i, M_j).}$$

At finite budget,

$$A_{ij}^{(B)} = \text{awa}_B(M_i, M_j).$$

The diagonal is

$$A_{ii} = \text{lev}(M_i).$$

The same object defines a directed awareness graph

$$G_{\text{awa}} = (V, E),$$

whose vertices are ITSs and whose directed edge

$$M_i \longrightarrow M_j$$

records the formal awareness of M_j available to M_i .

An edge can carry more than one quantity:

$$(\text{awa}(M_i, M_j), \text{awa}_B(M_i, M_j), H_{\text{awa}}(M_i, M_j, k)),$$

where

$$H_{\text{awa}}(M, N, k)$$

is the least verified cost required for M to attain awareness depth k about N .

This extends the general Proof Horizon idea: logical or formal availability and finite-budget accessibility are different quantities [2].

5 Awareness Preorders and Equivalence Classes

Fix a target N . Let

$$K_M(N)$$

denote the verifier-certified theory about N accessible to M .

Define

$$\boxed{M_1 \preceq_N M_2 \iff K_{M_1}(N) \subseteq K_{M_2}(N).}$$

This relation is reflexive and transitive and therefore forms a preorder.

The preorder induces an equivalence relation:

$$M_1 \sim_N M_2 \iff M_1 \preceq_N M_2 \wedge M_2 \preceq_N M_1.$$

Hence

$$M_1 \sim_N M_2 \iff K_{M_1}(N) = K_{M_2}(N).$$

The quotient

$$\mathcal{M}/\sim_N$$

groups together ITSs that may have radically different internal architectures but are equivalent with respect to the certified awareness they possess about N .

This is potentially important for security. A defender need not always reason separately about every conceivable attacker implementation. If multiple attackers lie in the same target-relative awareness class, they can sometimes be analyzed through one quotient class.

This also connects naturally with Quotient Hunter: quotienting is not confined to theorem states; it can potentially reduce spaces of agents, awareness states, or attack capabilities. A resource-sensitive version replaces $K_M(N)$ with $K_{M, \leq B}(N)$. Thus two ITSs may be extensionally awareness-equivalent while differing substantially at realistic budgets.

6 Controlled Formal Awareness

Traditional least privilege requires that an entity receive only the system resources and authorizations necessary to perform its function. NIST defines least privilege in precisely this operational spirit [8]. Controlled Formal Awareness proposes an inferential extension of that idea.

For every ordered pair M_i, M_j , define a lower awareness requirement

$$L_{ij}$$

and an upper permitted awareness bound

$$U_{ij}.$$

A desired operating region is

$$\boxed{L_{ij} \leq \text{awa}_B(M_i, M_j) \leq U_{ij}.}$$

The lower bound means that M_i understands enough about M_j to perform its legitimate function. The upper bound means that M_i does not acquire more formal inferential access to M_j than policy permits.

This can be called **inferential least privilege**.

A system may deny a component direct access to a protected database and nevertheless allow enough timing information, verifier responses, BANK provenance, or settlement outputs for the component to infer sensitive structure. CFA attempts to measure the resulting formal inferential capability rather than merely nominal file or API permissions.

An awareness intrusion can therefore be defined as an unauthorized transition across the permitted upper boundary:

$$\text{awa}_B(M_i, M_j) > U_{ij}.$$

More generally, intrusion may mean acquisition of a prohibited class of certified propositions about M_j . This complements conventional intrusion detection, which monitors and analyzes system or network events for signs of possible incidents [9].

7 Defensive Awareness Asymmetry

Because awareness is directed, asymmetry can itself be a security objective.

For a defensive system D and an untrusted worker X , one may want

$$\text{awa}(D, X) \gg \text{awa}(X, D).$$

The defender deeply models the worker, while the worker receives only limited information about the defender.

Define

$$\Delta(D, X) = \text{awa}(D, X) - \text{awa}(X, D)$$

when finite scalar levels are being compared. Positive Δ can represent **defensive awareness asymmetry**.

The broader architectural principle is not “maximize awareness everywhere.” It is

maximize authorized awareness and constrain unauthorized awareness.

In particular, strong self-awareness may be desirable on the diagonal A_{ii} , while sensitive off-diagonal entries A_{ij} , $i \neq j$, remain deliberately bounded.

This fits naturally with federated verified-memory architectures in which BANK scope, departmental separation, verifier sovereignty, provenance, and selective cross-department visibility are security variables [4].

8 Security Capability

Awareness is only one contributor to security. A separate quantity should measure actual defensive capability.

Let a threat model be

$$\Theta = (\mathcal{A}, I, T),$$

where $\mathcal{A}$ is a class of attacks, I is the protected invariant or security objective, and T records trust assumptions.

Define

$$\text{CapSec}_B(M; \Theta)$$

as the attacks in $\mathcal{A}$ against which M can preserve I , under budget B , according to the declared acceptance or certification criteria.

Then define the security-capability preorder

$$M_1 \preceq_{\text{sec}} M_2 \iff \text{CapSec}(M_1; \Theta) \subseteq \text{CapSec}(M_2; \Theta).$$

Again this induces equivalence classes.

The natural upper bound is not necessarily “all imaginable attacks.” Some attacks are undefeatable once the trust assumptions themselves are violated. Let

$$\mathcal{A}_{\text{def}}$$

denote the attacks defensible under Θ .

Then maximal capable security satisfies

$$\text{CapSec}(M; \Theta) = \mathcal{A}_{\text{def}}.$$

Thus an ITS is maximally security-capable relative to a threat model when it can successfully defend against every attack that is defensible under that model.

9 Does Greater Self-Awareness Improve Security Capability?

Suppose M_k denotes a version of a defender having self-awareness depth k .

Under ideal conditions—additional awareness is sound, private, verifier-safe, nonleaking, and can simply be ignored when unhelpful—one expects

$$k \leq j \implies \text{CapSec}(M_k) \subseteq \text{CapSec}(M_j).$$

This suggests a provisional **Awareness Monotonicity Principle**:

Cost-free, sound, private additional formal self-awareness cannot reduce optimal defensive capability.

Under that idealization,

$$\text{lev}(M) = \omega$$

is a natural upper bound for the self-awareness coordinate of a maximally capable defender.

This does not establish that greater awareness is always safe in a real architecture. More modeling machinery may create interfaces, memory, side channels, or vulnerabilities. Thus the principle concerns an idealized capability upper bound, not an unconditional empirical security theorem.

10 Why ω -Awareness Is Not Logical Omniscience

Even if

$$\text{lev}(M) = \omega,$$

this does not imply that M knows every true proposition about itself. Arbitrary finite reflective depth is different from semantic completeness.

A sufficiently strong, consistent effective formal system cannot in general prove its own consistency under the standard hypotheses of Gödel’s second incompleteness theorem. Löb-style reflection results impose further restrictions on unrestricted self-certification [11].

Therefore

$$\text{lev}(M) = \omega$$

should concern unbounded finite **formal modeling depth**, not unrestricted proof of the complete soundness, consistency, or future behavior of M .

11 Nonstandard Analysis as a Source of Bounds

Nonstandard analysis should enter this theory as a source of bounds, not as a claim that DATA-MIND literally becomes a hyperfinite physical computer.

Let

$$H \in {}^*\mathbb{N}$$

be an unlimited hyperinteger, so that

$$H > n \quad \text{for every standard } n \in \mathbb{N}.$$

Nonstandard analysis supplies ideal elements of this kind and, under the appropriate formal conditions, the transfer principle permits standard finite statements of the appropriate logical form to be transferred to their nonstandard extensions [12].

Suppose a budget-indexed quantity $Q(B)$ is monotone:

$$B_1 \leq B_2 \implies Q(B_1) \preceq Q(B_2).$$

Then its nonstandard extension permits comparison at an unlimited bound:

$${}^*Q(n) \preceq {}^*Q(H)$$

for every standard finite n , whenever the transferred statement is legitimate.

This can be applied schematically to

$$\text{awa}_B(M, N), \quad \text{lev}_B(M), \quad \text{CapSec}_B(M),$$

or proof-horizon quantities.

The correct interpretation is

$$\boxed{\text{NSA supplies an ideal upper comparison regime for finite behavior.}}$$

It does not mean the deployed ITS performs literally infinitely many operations.

This role is consistent with earlier work connecting nonstandard analysis, finite proof limits, and certified presentation trading [5].

12 The Frozen Ocean and the Proposed Ocean Moat

The Frozen Ocean construction supplies a different notion of depth. In the benchmark family $F(n)$, the relevant quantity is calibrated so that the shortest proof has length

$$\boxed{L^*(F(n)) = n.}$$

Thus n is literal minimum proof depth. The Frozen Ocean program uses controlled theorem depth and density as objective quantities for ATP measurement [1].

This Ocean depth should **not** be identified with awareness depth.

An Ocean-style security mechanism might interpose a deep proof problem between an ITS query and information release. Schematically,

$$q \longmapsto F_q(n) \longmapsto \text{verified settlement} \longmapsto \text{response}.$$

If n is large, every accepted proof may be forced to have substantial depth.

However, this does not by itself create strong security. A known proof route can be easy to find even if its certificate necessarily contains n steps. Therefore

minimum proof length $\neq$ proof-search difficulty.

A serious Ocean Moat would need to distinguish at least proof depth from discovery difficulty and verification cost.

Thus the Ocean Moat should currently be treated as a **candidate enforcement mechanism for Controlled Formal Awareness**, not as part of CFA’s foundation. It may be useful if acquiring information that would increase unauthorized awareness requires crossing deliberately deep and search-resistant theorem structures.

13 Public Architecture and the Moat

A defense whose value disappears as soon as the adversary knows the mechanism is weak. Saltzer and Schroeder’s classical open-design principle states that protection should not depend on secrecy of the design; their framework also includes least privilege, complete mediation, separation of privilege, and related design principles [10].

Therefore any eventual Ocean Moat should ideally satisfy a public-architecture condition:

Knowledge that the moat exists, and preferably knowledge of its construction method, should not by itself collapse the security guarantee.

This is one reason proof depth alone is insufficient. If disclosure immediately reveals a deterministic n -step route, the Ocean remains deep in certificate length but shallow in search.

14 Federation as Controlled Awareness Topology

A federated DATA-MIND or AMLD can now be described simultaneously as

a settlement architecture,

a verified-memory topology,

and

an awareness graph.

BANK visibility, verifier outputs, department coupling, provenance, Sentinel decisions, and controller communication can all change edges in G_{awa} without necessarily changing the verifier’s acceptance relation. The federated-memory security motivation is developed in AMLD Security 4 [4].

This suggests a new security-design problem:

Choose the federation topology so that required awareness is reachable while prohibited awareness remains bounded.

In matrix form,

$$L_{ij} \leq A_{ij}^{(B)} \leq U_{ij}.$$

In graph form, it is a constrained directed-information problem. In preorder form, it is a problem of preventing an unauthorized agent from moving upward into stronger target-awareness equivalence classes.

These are three views of the same underlying structure.

15 Working Synthesis

The central proposal is

Controlled Formal Awareness Security.

It studies the permitted and prohibited formal inferential relationships among ITSs.

Its primitive relation is

$$\text{awa}(M, N).$$

Its self-awareness diagonal is

$$\text{lev}(M) = \text{awa}(M, M).$$

Its finite operational form is

$$\text{awa}_B(M, N).$$

Its upper-bound interpretation is

$$\text{awa}(M, N) = \omega$$

when finite achievable awareness depth is unbounded.

Its system-wide structures are the awareness matrix and directed awareness graph. Its order theory arises from inclusion of certified awareness theories, producing preorders and quotient equivalence classes. Its security rule is controlled inferential privilege:

$$L_{ij} \leq \text{awa}_B(M_i, M_j) \leq U_{ij}.$$

Its capability objective is

$$\text{CapSec}(M) = \mathcal{A}_{\text{def}}$$

for a maximally capable defender relative to a declared threat model.

Its ideal defensive tendency is high authorized self- and supervisory awareness together with restricted unauthorized cross-machine awareness. Nonstandard analysis supplies upper comparison bounds for finite-budget behavior. Frozen Ocean depth supplies an independent calibrated notion of proof depth that may become a security mechanism, but proof depth must not be confused with awareness or with search hardness.

The resulting picture is therefore not simply “more awareness means more security.” It is

Security is the controlled geometry of who can formally know what about whom, at what depth, under what budget, and with what defensive consequence.

Status of This Draft

This document is a working theoretical synthesis. The definitions of awa , awa_B , the awareness preorder, Controlled Formal Awareness, and the security-capability preorder are proposals developed in this draft unless explicitly attributed to earlier work. Claims drawn from Tenneson’s prior corpus are cited to the corresponding public GitHub files. Standard external references are used for least privilege, intrusion detection, open design, incompleteness, and nonstandard-analysis background. The trust-anchor verifier remains sovereign: reflective, learned, federated, traded, or awareness-based mechanisms may alter search, visibility, modeling, or resource allocation, but they must not silently change what counts as an accepted certificate.

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